EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Group 1 Elements; Alkali Metals, Electronic Configuration, Atomic and Ionic Radii, Ionisation Enthalpy, Hydration Enthalpy, Physical Properties

1. Alkali metals give colour in Bunsen flame due to:

(A) Low electronegativity (B) One electron in outer most orbit

(C) Smaller atomic radii (D) Low ionisation energy

CSB132

2. Choose the correct order of thermal Stability for IA element oxides.

(A) $Li_2O > Na_2O > K_2O > Rb_2O > Cs_2O$ (B) $Li_2O < Na_2O < K_2O < Rb_2O < Cs_2O$

(C) $Na_2O > k_2O > Li_2O > Rb_2O > Cs_2O$ (D) $Na_2O < k_2O > Li_2O < Rb_2O < Cs_2O$

CSB129

3. Select the correct statement.

(A) Among the alkali metals, only lithium reacts with nitrogen directly at room temperature to form nitride.

(B) Among the alkali metal carbonates, Li_2CO_3 has the lowest thermal stability.

(C) Among the alkali metal hydroxide, CsOH has the highest solubility in water.

(D) All of these.

CSB147

Rankers Academy JEE

Chemical properties of Alkali metals (Reactivity towards Air, Water, Halogens, Dihydrogen, Reducing Nature, Solutions in Liquid NH_3)

4. What happens when sodium metal is heated in excess of dry air containing carbon dioxide gas?

CSB131

(A) Na_2O_2 is formed. (B) Na_2O is formed. (C) Na_2CO_3 is formed. (D) Na_3N is formed.

5. Which of the following about solution of alkali metals in liquid ammonia is correct?

(A) The solutions have strong oxidizing properties.

(B) Both the dilute solution as well as concentrated solution are paramagntic in nature.

(C) Colour of the solution is attributed to charge transfer spectrum.

(D) None of these.

CSB133

6. Solution of sodium metal in liquid ammonia is a strong reducing agent due to presence of

(A) solvated sodium ions (B) solvated hydrogen ions

(C) sodium atoms or sodium hydroxide (D) solvated electrons

CSB119

7. Nitrate of an element of alkali metal group, decomposes on heating, gives brown colour gas. Nitrate and brown colour gas are respectively:-

(A) $NaNO_3$ and NO (B) $LiNO_3$ and NO_2

(C) KNO_3 and NH_3 (D) $NaNO_3$ and NO_2

CSB139

For More Join: @IITJEE_Advanced

8. Consider the following statements ;

S_1 : Among alkali metal halides lithium iodide is the most covalent in nature.

S_2 : Potassium has greater photoelectric work function than sodium.

S_3 : The blue solution of alkali metals in liquid ammonia is stable at room temperature, where ammonia is still a liquid, in the presence of Fe.

S_4 : The melting and boiling points of alkali metal halides always follow the trend : chloride > fluoride > bromide > iodide.

and arrange in the order of true/false.

(A) T F F F (B) T T F F

(C) T F T T (D) T T T F

CSB135

General Characteristic of Compounds of Alkali Metals (Oxides and Hydroxides, Halides, Salts of Oxo-acids)

9. Carbon can be oxidised to CO_2 while heating its powder with:

(A) SO_2 (B) KNO_3 (C) K_2CO_3 (D) FeS_2

CSB137

Some Important Compounds of Sodium ($Na_2CO_3.10H_2O$; NaOH; $NaHCO_3$;NaCl) and their uses

10. (Milky Cloud) C $\xleftarrow{CO_2}$ A + Na_2CO_3 $\longrightarrow$ B + C

The chemical formulae of A and B are

(A) NaOH and $Ca(OH)_2$ (B) $Ca(OH)_2$ and NaOH

(C) NaOH and CaO (D) CaO and $Ca(OH)_2$

CSB116

Rankers Academy JEE

Group 2 Elements; Alkaline Earth Metals, Electronic Configuration, Atomic and Ionic Radii, Ionisation Enthalpy, Hydration Enthalpy, Physical Properties

11. Which option is correct for density of alkaline earth metals.

(A) Ca > Mg (B) Sr < Ba

(C) Mg > Be (D) Ba < Sr

CSB110

12. Choose the correct order for decreasing order of melting point

(A) Be > Mg > Ca > Sr > Ba (B) Be < Mg < Ca < Sr < Ba

(C) Be > Ca > Sr > Ba > Mg (D) Be > Ca > Mg > Sr > Ba

CSB121

13. Choose the incorrect option for melting point.

(A) NaF < MgF_2 (B) NaF < MgO

(C) BeO < BeF_2 (D) NaF > KF

CSB127

14. The order of thermal stability of carbonates of IIA group is.

(A) $BaCO_3 > SrCO_3 > CaCO_3 > MgCO_3$ (B) $MgCO_3 > CaCO_3 > SrCO_3 > BaCO_3$

(C) $CaCO_3 > SrCO_3 > BaCO_3 > MgCO_3$ (D) $MgCO_3 = CaCO_3 > SrCO_3 = BaCO_3$

CSB128

15. Which one of the following compounds are readily soluble in water?

(A) $MgSO_4$ (B) $SrSO_4$ (C) $BaSO_4$ (D) None

CSB142

For More Join: @IITJEE_Advanced

16. Which of the following is incorrect statement:
(A) Solubilities of alkaline earth metal fluorides and hydroxides generally increase down the group.
(B) Hydration energies of alkali metal halides decrease down the group with increase in size of cations.
(C) Mg^{2+} ion is bigger than Li^{+}.
(D) $BeCl_2$ is easily hydrolysed in water.

CSB146

17. Which of the following properties show a reverse trend in moving down the group of alkali & alkaline earth metals ?
(A) Solubility of hydroxides (B) Solubility of oxides
(C) Solubility of fluorides (D) Solubility of carbonates

CSB151

Chemical properties of Alkaline earth metals (Reactivity towards Air, Water, Halogens, Acids, Reducing Nature, Solution in Liquid NH_3)

18. Which salt hydrolyses to a minimum extent?
(A) $Mg(NO_3)_2$ (B) $Be(NO_3)_2$ (C) $Ca(NO_3)_2$ (D) $Ba(NO_3)_2$

CSB138

19. An alkaline earth metal (M) gives a salt with chlorine, which is soluble in water at room temperature. It also forms an insoluble sulphate whose mixture with a sulphide of a transition metal is called 'lithopone' -a white pigment. Metal M is.
(A) Ca (B) Mg (C) Ba (D) Sr

CSB125

Rankers Academy JEE

20. When ionic nitrides react with water, the products are :
(A) acidic solution and hydrogen gas. (B) acidic solution and ammonia gas.
(C) basic solution and ammonia gas. (D) basic solution and hydrogen gas.

CSB144

21. A metal (M) burns with dazzling brilliance in air above 1000°C to give a white powder. The white powder reacts with water to form a white precipitate (P) and a colourless gas (G) with a characteristic smell. The metal (M) dissolves in conc. NaOH to liberate another gas(H). (H) may also be obtained on heating (G). Then:
(A) M = Mg (B) M = Be (C) P = $Ca(OH)_2$ (D) G = O_3

CSB145

22. $X + C + Cl_2 \xrightarrow[\text{of about } 1000\,K]{\text{High temperature}} Y + CO$;

$Y + 2H_2O \uparrow Z + 2HCl$

Compound Y is found in polymeric chain structure in solid phase and is an electron deficient molecule. Y must be
(A) BeO (B) $BeCl_2$
(C) BeH_2 (D) $AlCl_3$

CSB123

23. $BeCl_2 + LiAlH_4 \longrightarrow X + LiCl + AlCl_3$
(A) X is LiH (B) X is BeH_2
(C) X is $BeCl_2 \cdot 2H_2O$ (D) None

CSB124

For More Join: @IITJEE_Advanced

24. Select the incorrect statement :
(A) Solutions of alkali metals in liquid ammonia are a good reducing agent because they contain free or solvated electrons.
(B) The crystalline salts of alkaline earth metals contain more water of crystallisation than the corresponding alkali metal salts.
(C) Atoms of alkaline earth metals have smaller size and more nuclear charge than alkali metal atoms of same period.
(D) All alkali metal halides form hydrates.

CSB134

25. A metal which is soluble in both water and liquid NH_3 separately -
(A) Cr (B) Mn (C) Ba (D) Al

CSB109

Uses of Alkaline Earth Metals

26. A doctor by mistake administers a dilute $Ba(NO_3)_2$ solution to a patient for radiagraphic investigations. Which of the following should be the best to prevent the absorption of soluble Barium and subsequent Barium poisoning.
(A) NaCl (B) Na_2SO_4 (C) Na_2CO_3 (D) NH_4Cl

CSB148

General Characteristic of Compounds of Alkaline Earth Metals (Oxides and Hydroxides, Halides, Salts of Oxo-acids(CO_3^{2-}, SO_4^{2-},NO_3^-))

27. A metal M readily forms water soluble sulphate, and water insoluble hydroxide $M(OH)_2$. Its oxide MO is amphoteric, hard and having high melting point. The alkaline earth metal M must be
(A) Mg (B) Be (C) Ca (D) Sr

CSB130

Rankers Academy JEE

Some Important Compounds of Calcium (CaO, $Ca(OH)_2$, $CaCO_3$, $CaSO_4.\frac{1}{2}H_2O$)

28. Dry Slaked lime powder is.
(A) Dry form of $Ca(OH)_2$ (B) Soln. of $Ca(OH)_2$
(C) Dry form of $CaCO_3$ (D) Dry form of CaO

CSB114

29. $Y \xleftarrow{\Delta,205°C} CaSO_4\cdot 2H_2O \xrightarrow{\Delta,120°C} X$. X and Y are respectively
(A) plaster of paris, dead burnt plaster (B) dead burnt plaster, plaster of paris
(C) CaO and plaster of paris (D) plaster of paris, mixture of gases

CSB115

Cement

30. The purpose of addition of gypsum in the cement is -
(A) To slow down the process of setting of the cement
(B) To fasten the process of setting of the cement
(C) Not to affect the process of setting of the cement by any means
(D) None of these

CSB117

31. Which of the following statement is **incorrect**:
(A) K_2CO_3 can also be prepared by Solvay process
(B) Li_2CO_3 does not exist in solid form
(C) Be is inert towards reaction with water
(D) Gypsum slow down the process of setting of cement

CSB140

For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

Group 1 Elements; Alkali Metals, Electronic Configuration, Atomic and Ionic Radii, Ionisation Enthalpy, Hydration Enthalpy, Physical Properties

1. Select correct statement(s):

(A) Stability of peroxides and superoxides of alkali metals increases with increase in size of the cation.

(B) Increase in stability in (A) is due to stabilisation of large anions by larger cations through lattice energy effects.

(C) The low solubility of LiF is due to its high lattice energy whereas low solubility of CsI is due to smaller hydration energy.

(D) NaOH is not deliquescent.

CSB152

2. Which of the following are correctly matched?

(A) Basic strength $Cs_2O < Rb_2O < K_2O < Na_2O < Li_2O$

(B) Stability of peroxides $Na_2O_2 < K_2O_2 < Rb_2O_2 < Cs_2O_2$

(C) Stability of bicarbonates $LiHCO_3 < NaHCO_3 < KHCO_3 < RbHCO_3 < CsHCO_3$

(D) Thermal stability of hydrides $CsH < RbH < KH < NaH < LiH$

CSB153

General Characteristic of Compounds of Alkali Metals (Oxides and Hydroxides, Halides, Salts of Oxo-acids)

Bankers Academy JEE

3. Which of the following statement(s) is/are correct?

(A) Milk of lime is a suspension of $Ca(OH)_2$ in water.

(B) Lime water is a clear solution of $Ca(OH)_2$ in water.

(C) Baryta water is a clear solution of $Ba(OH)_2$.

(D) Nitrolim is the mixture of $CaCN_2$ and carbon.

CSB154

Group 2 Elements; Alkaline Earth Metals, Electronic Configuration, Atomic and Ionic Radii, Ionisation Enthalpy, Hydration Enthalpy, Physical Properties

4. The set representing the incorrect order of first ionisation potential is :

(A) B > C > N (B) Ge > Si > C

(C) K < Na < Li (D) Be > Mg > Ca

CSB156

5. Which is/are true statement(s)?

(A) The heats of hydration of the dipositive alkaline earth metal ions decreases with an increase in their ionic size.

(B) Hydration of alkali metal ion is less than that of II A ion of the same period.

(C) Alkaline earth metal ions, because of their much larger charge to radius ratio, exert a much stronger electrostatic attraction on the oxygen of water molecule surrounding them.

(D) None.

CSB158

For More Join: @IITJEE_Advanced

EXERCISE (S)

Part-1 : JEE Main (Numerical Value type)

Group 1 Elements; Alkali Metals, Electronic Configuration, Atomic and Ionic Radii, Ionisation Enthalpy, Hydration Enthalpy, Physical Properties

1. How many of the following are correctly matched?

Element	**Colour in flame test**
K	Violet/Lilac
Na	Yellow
Be	Crimson red
Ca	Brick red
Sr	Apple green
Mg	No colour
Rb	Red violet
Cs	Blue
Li	Crimson red

CSB161

Chemical properties of Alkali metals (Reactivity towards Air, Water, Halogens, Dihydrogen, Reducing Nature, Solutions in Liquid NH_3)

2. For alkali metal M :

$M_2O + H_2O \rightarrow x$

$M_2O_2 + H_2O \rightarrow x + y$

$MO_2 + H_2O \rightarrow x + y + z$

Sum of the number of atoms present in one molecule each of x, y, z.

Rankers Academy JEE

CSB167

General Characteristic of Compounds of Alkali Metals (Oxides and Hydroxides, Halides, Salts of Oxo-acids)

3. Select the number of reactions which is feasible :

(i) $K_2MnO_4 \xrightarrow{H^+} KMnO_4 + MnO_4$

(ii) $Cl_2 + NaOH \longrightarrow NaClO + NaCl + H_2O$

(iii) $F_2 + NaOH \longrightarrow NaFO + NaF + H_2O$

(iv) $Br_2 + KOH \longrightarrow KBrO + KBr + H_2O$

CSB159

4. The number of pair of compounds which cannot exist together in aqueous solution is :-

(i) NaH_2PO_4 and $NaHCO_3$ (ii) Na_2CO_3 and $NaHCO_3$

(iii) NaOH and NaH_2PO_2 (iv) $NaHCO_3$ and NaOH

CSB160

Some Important Compounds of Sodium ($Na_2CO_3.10H_2O$; NaOH;$NaHCO_3$;NaCl) and their uses

5. Molecular formula of Glauber's salt is $Na_2SO_4.xH_2O$. The value of x is____.

CSB163

For More Join: @IITJEE_Advanced

6. The byproduct of Solvay process reacts with Na_2CO_3 to form a compound x, which on heating decomposes to give y, y is absorbed by KO_2. The number of atoms per molecule of y is .....

CSB164

Group 2 Elements; Alkaline Earth Metals, Electronic Configuration, Atomic and Ionic Radii, Ionisation Enthalpy, Hydration Enthalpy, Physical Properties

7. Among the following how many orders of property are correct.

(i) LiH < NaH < KH < RbH < CsH (Thermal stability)

(ii) $BeCO_3 < MgCO_3 < CaCO_3 < SrCO_3$ (Solubility)

(iii) $Na_2SO_4 < K_2SO_4 < Rb_2SO_4 < Cs_2SO_4$ (Solubility)

(iv) 1 & 3 both

CSB197

Chemical properties of Alkaline earth metals (Reactivity towards Air, Water, Halogens, Acids, Reducing Nature, Solution in Liquid NH_3)

8. How many nitrate groups are present in 1 molecule of Basic beryllium nitrate?

CSB166

9. $A + B + H_2O \longrightarrow (NH_4)HCO_3$

$NH_4HCO_3 + NaCl \longrightarrow C + NH_4Cl$

$NH_4Cl + D \longrightarrow 2NH_3 + 2H_2O + CaCl_2$

Sum of the atoms present in one molecule each of A, B, C and D.

CSB170

10. How Many of the following reaction is not correct

(i) $Cl_2 + NaOH \xrightarrow{Hot/Conc.} NaCl + NaClO_3 + H_2O$

(ii) $P_4 + NaOH \longrightarrow PH_3 + NaH_2PO_2$

(iii) $S + NaOH \longrightarrow Na_2S_2O_3 + Na_2S + H_2O$

(iv) $Si + NaOH \longrightarrow Na_4SiO_4 + H_2$

Rankers Academy JEE

CSB171

11. How many statements are true

(a) $CaSO_4.2H_2O \xrightarrow{393K} CaSO_4 \frac{1}{2} H_2O$

(b) $NaOH(hot\ \&\ con.) + Br_2 \longrightarrow NaBr + NaBrO_4 + H_2O$

(c) $Mg + N_2 \longrightarrow Mg_3N_2 \xrightarrow{+H_2O} NH_3$ (gas)

(d) $Be + C \longrightarrow Be_2C \xrightarrow{+H_2O} CH_4$

CSB172

Part-2 : JEE Advanced (Decimal type)

Chemical properties of Alkali metals (Reactivity towards Air, Water, Halogens, Dihydrogen, Reducing Nature, Solutions in Liquid NH_3)

12. How many of the following are correctly matched:

Hydration energy :	$Li^+ > Na^+ > K^+ > Rb^+ > Cs^+$
Mobility of ions (aq) :	$Li^+ > Na^+ > K^+ > Rb^+ > Cs^+$
Density :	Li < Na < K < Rb < Cs
Reaction with N_2 :	Li < Na < Rb < Cs < K
Reducing nature of gas phase :	Li < Na < K < Rb < Cs
Reducing nature in aq. phase :	Li > Na > K > Rb > Cs

CSB198

For More Join: @IITJEE_Advanced

Some Important Compounds of Sodium ($Na_2CO_3.10H_2O$; NaOH; $NaHCO_3$; NaCl) and their uses

13. How many of the following disproportionate(s) on heating with sodium hydroxide?
(Give your final answer by dividing 2)
(i) P_4 (ii) S_8 (iii) Cl_2 (iv) B.

CSB200

Chemical properties of Alkaline earth metals (Reactivity towards Air, Water, Halogens, Acids, Reducing Nature, Solution in Liquid NH_3)

14. How many of the following statement is/are correct?
(a) Solvay process is used for manufacturing sodium carbonate.
(b) $CaCl_2$ is obtained as by product in Solvay process.
(c) NH_3 can be recovered in above process.
(d) $MgSO_4.7H_2O$ is epsom salt.
(e) On hydrolysis of Na_2CO_3, we get an acidic solution due to the formation of H_2CO_3.
(f) K_2CO_3 can also be prepared by Solvay process.
(g) $CaCO_3$ can be obtained by passing excess of CO_2 through lime water.

CSB169

Some Important Compounds of Calcium (CaO, $Ca(OH)_2$, $CaCO_3$, $CaSO_4.1/2H_2O$)

15. When gypsum is heated at 393 K, the compound formed is $CaSO_4.xH_2O$. Value of x is.....

CSB165

Rankers Academy JEE

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. Match **List-I** with **List-II.**

List-I **Alkali Metal**	**List-II** **Emission Wavelength in nm**
(A) Li	(I) 589.2
(B) Na	(II) 455.5
(C) Rb	(III) 670.8
(D) Cs	(IV) 780.0

Choose the **correct** Answer from the options given below:

[JEE (Main) 2024]

(1) (A)-(I), (B)-(IV), (C)-(III), (D)-(II) (2) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)
(3) (A)-(IV), (B)-(II), (C)-(I), (D)-(III) (4) (A)-(II), (B)-(IV), (C)-(III), (D)-(I)

CSB201

2. Alkali metal from the following with least melting point is: **[JEE (Main) 2023]**

(1) Rb (2) K (3) Na (4) Cs

CSB189

3. Which one among the following metals is the weakest reducing agent?

[JEE (Main) 2023]

(1) K (2) Rb (3) Na (4) Li

CSB177

Pankers Academy JEE

4. Identify the correct statements about alkali metals.

A. The order of standard reduction potential ($M^+ | M$) for alkali metal ions is Na > Rb > Li.
B. CsI is highly soluble in water.
C. Lithium carbonate is highly stable to heat.
D. Potassium dissolved in concentrated liquid ammonia is blue in colour and paramagnetic.
E. All the alkali metal hydrides are ionic solids.

Choose the correct answer from the options given below **[JEE (Main) 2023]**

(1) A, B, D only (2) C and E only
(3) A and E only (4) A, B and E only

CSB175

5. Which of the following reaction is correct ? **[JEE (Main) 2023]**

(1) $2LiNO_3 \xrightarrow{\Delta} 2LiNO_2 + O_2$ (2) $4LiNO_3 \xrightarrow{\Delta} 2Li_2O + 2N_2O_4 + O_2$

(3) $4LiNO_3 \xrightarrow{\Delta} 2Li_2O + 4NO_2 + O_2$ (4) $2LiNO_3 \xrightarrow{\Delta} 2Li + 2NO_2 + O_2$

CSB179

6. Number of water molecules in washing soda and soda ash respectively are: **[JEE (Main) 2023]**

(1) 10 and 1 (2) 1 and 10 (3) 1 and 0 (4) 10 and 0

CSB188

7. Ion having highest hydration enthalpy among the given alkaline earth metal ions is:

[JEE (Main) 2023]

(1) Be^{2+} (2) Ba^{2+} (3) Sr^{2+} (4) Ca^{2+}

CSB178

For More Join: @IITJEE_Advanced

8. Given below are two statements:
Statement I: Lithium and Magnesium do not form superoxide
Statement II: The ionic radius of Li^+ is larger than ionic radius of Mg^{2+}
In the light of the above statements, choose the most appropriate answer from the options given below: **[JEE (Main) 2023]**
(1) Statement I is incorrect but Statement II is correct
(2) Statement I is correct but Statement II is incorrect
(3) Both Statement I and Statement II are correct
(4) Both Statement I and Statement II are incorrect

CSB185

9. $Mg(NO_3)_2 \cdot XH_2O$ and $Ba(NO_3)_2 \cdot YH_2O$, represent formula of the crystalline forms of nitrate salts. Sum of X and Y is ____ **[JEE (Main) 2023]**

CSB190

10. Given below are two statements : One is labelled as Assertion A and the other is labelled as Reason R :
Assertion (A) : $BeCl_2$ and $MgCl_2$ produce characteristic flame.
Reason (R) : The excitation energy is high in $BeCl_2$ and $MgCl_2$
In the light of the above statements, choose the correct answer from the option given below : **[JEE (Main) 2023]**
(1) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)
(2) (A) is false but (R) is true
(3) Both (A) and (R) are true and (R) is the correct explanation of (A)
(4) (A) is true but (R) is false.

CSB180

11. Choose the correct statement(s):
A. Beryllium oxide is purely acidic in nature.
B. Beryllium carbonate is kept in the atmosphere of CO_2.
C. Beryllium sulphate is readily soluble in water.
D. Beryllium shows anomalous behaviour.
Choose the correct answer from the options given below: **[JEE (Main) 2023]**
(1) A, B and C only (2) B, C and D only (3) A and B only (4) A only

Rankers Academy JEE

CSB182

12. $Be(OH)_2$ react with $Sr(OH)_2$ to yield an ionic salt. Choose the incorrect option related to this reaction from the following: **[JEE (Main) 2023]**
(1) Be is tetrahedrally coordinated in the ionic salt.
(2) The reaction is an example of acid - base neutralization reaction.
(3) Both Sr and Be elements are present in the ionic salt.
(4) The elements Be is present in the cationic part of the ionic salt.

CSB192

13. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : Gypsum is used for making fireproof wall boards.
Reason (R) : Gypsum is unstable at high temperatures.
In the light of the above statements, choose the **correct** answer from the options given below : **[JEE (Main) 2023]**
(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**.
(2) **(A)** is correct but **(R)** is not correct.
(3) **(A)** is not correct but **(R)** is correct.
(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**. **CSB183**

For More Join: @IITJEE_Advanced

14. Lime reacts exothermally with water to give 'A' which has low solubility in water. Aqueous solution of 'A' is often used for the test of CO_2, a test in which insoluble B is formed. If B is further reacted with CO_2 then soluble compound is formed 'A' is

[JEE (Main) 2023]

(1) Quick lime (2) Slaked lime (3) Lime water (4) White lime

CSB186

15. The setting time of Cement is increased by adding **[JEE (Main) 2023]**

(1) Clay (2) Silica (3) Limestone (4) Gypsum

CSB184

16. Match List I with List II.

List I		List II	
A	Chlorophyll	I	Na_2CO_3
B	Soda ash	II	$CaSO_4$
C	Dentistry, Ornamental work	III	Mg^{2+}
D	Used in white washing	IV	$Ca(OH)_2$

Choose the correct answer from the options given below : **[JEE (Main) 2023]**

(1) A – III, B – I, C – II, D – IV (2) A – II, B – I, C – III, D – IV

(3) A – III, B – IV, C – I, D – II (4) A – II, B – III, C – IV, D – I

CSB174

17. Order of Covalent bond;

A. KF > KI; LiF > KF

B. KF < KI; LiF > KF

C. $SnCl_4 > SnCl_2$; CuCl > NaCl

D. LiF > KF; CuCl < NaCl

E. KF < KI; CuCl > NaCl **[JEE (Main) 2023]**

(1) C, E only (2) B, C only (3) B, C, E only (4) A, B only

CSB173

Pankers Academy JEE

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Advanced) PYQ

1. The following compounds have been arranged in order of their increasing thermal stabilities. Identify the correct order.
K_2CO_3(I) $MgCO_3$(II) $CaCO_3$(III) $BeCO_3$(IV) **[JEE (Advanced) 1996]**
(A) I < II < III < IV (B) IV < II < III < I (C) IV < II < I < III (D) II < IV < III < I
CSB073

2. Read the following statement and explanation and answer as per the options given below :
Statement-1 : The alkali metals can form ionic hydrides which contain the hydride ion $H^{\ominus}$.
Statement-2 : The alkali metals have low electronegativity ; their hydrides conduct electricity when fused and liberate hydrogen at the anode. **[JEE (Advanced) 1994]**
(A) Both 1 and 2 are true and 2 is the correct explanation of 1.
(B) Both 1 and 2 are true but 2 is not the correct explanation of 1.
(C) 1 is true but 2 is false.
(D) 1 is false but 2 is true.
CSB072

3. This questions contains statement-1 (assertion) and statement-2 (reason) and has 4 choices (a), (b), (c) and (d) out of which only one is correct.
Statement-1 : Alkali metals dissolve in liquid ammonia to give blue solutions. because.
Statement-2 : Alkali metals is liquid ammonia give solvated species of the type $[M(NH_3)_n]^+$ (M = alkali metals). **[JEE (Advanced) 2007]**
(A) Both 1 and 2 are true and 2 is the correct explanation of 1.
(B) Both 1 and 2 are true but 2 is not the correct explanation of 1.
(C) 1 is true but 2 is false.
(D) 1 is false but 2 is true.
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CSB077

4. Highly pure dilute solution of sodium in liquid ammonia - **[JEE (Advanced) 1998]**
(A) shows blue colour (B) exhibits electrical conductivity
(C) produces sodium amide (D) produces hydrogen gas
CSB075

5. List-I contains compounds and List-II contains reaction **[JEE (Advanced) 2022]**

List-I	List-II
(I) H_2O_2	(P) $Mg(HCO_3)_2 + Ca(OH)_2$
(II) $Mg(OH)_2$	(Q) $BaO_2 + H_2SO_4 \rightarrow$
(III) $BaCl_2$	(R) $Ca(OH)_2 + MgCl_2$
(IV) $CaCO_3$	(S) $BaO_2 + HCl \rightarrow$
	(T) $Ca(HCO_3)_2 + Ca(OH)_2 \rightarrow$

Match each compound in List- I with its formation reaction(s) in List-II, and choose the correct option
(A) I $\rightarrow$ Q; II $\rightarrow$ P; III $\rightarrow$ S; IV $\rightarrow$ R (B) I $\rightarrow$ T; II $\rightarrow$ P; III $\rightarrow$ Q; IV $\rightarrow$ R
(C) I $\rightarrow$ T; II $\rightarrow$ R; III $\rightarrow$ Q; IV $\rightarrow$ P (D) I $\rightarrow$ Q; II $\rightarrow$ R; III $\rightarrow$ S; IV $\rightarrow$ P
CSB193

6. Which of the following liberates O_2 upon hydrolysis? **[JEE (Advanced) 2020]**
(A) Pb_3O_4 (B) KO_2 (C) Na_2O_2 (D) Li_2O_2
CSB202

7. The species that do not contain peroxide linkage are - **[JEE (Advanced) 1992]**
(A) PbO_2 (B) H_2O_2 (C) SrO_2 (D) BaO_2
CSB071

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ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	A	D	C	D	D	B	A	B	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	C	C	A	A	C	D	D	C	C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	B	B	D	C	B	B	A	A	A
Que.	31									
Ans.	A									

EXERCISE (O-2)

Que.	1	2	3	4	5	
Ans.	ABC	BCD	ABCD	CD	ABC	

EXERCISE (S)

Part-1 : JEE Main (Integer type)

1. 7 (All except Be & Sr) **2.** 9 **3.** 3 **4.** 1
5. 10 **6.** 3 **7.** 1 **8.** 6 **9.** 18
10. 1 **11.** 3 **12.** 2.00 **13.** 1.5
14. (a), (b), (c), (d) are true. **15.** 0.50

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EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	4	3	3	3	4	1	3	6	2
Que.	11	12	13	14	15	16	17			
Ans.	2	4	1	2	4	1	3			

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	
Ans.	B	A	B	AB	D	B	A	

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Important Notes

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